



Project Overview:

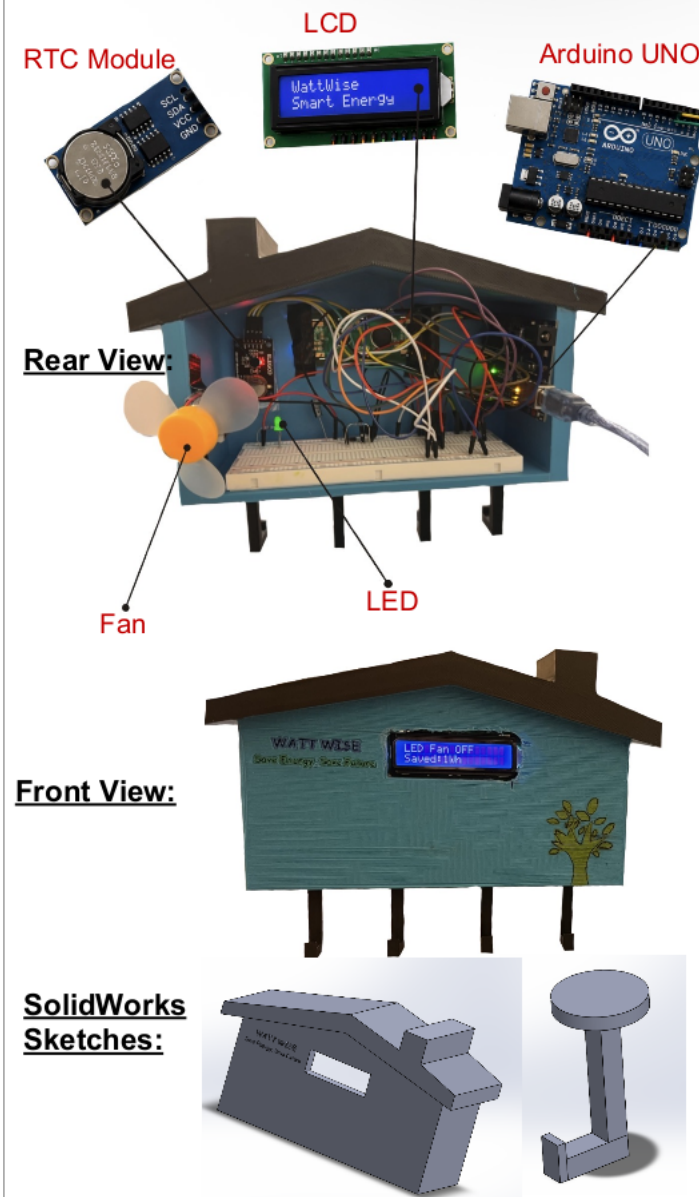
WattWise is a smart home energy management system that helps users reduce energy consumption. The device tracks energy usage throughout the day, displays the amount of energy saved in watts, and allows appliances and lights to be automated with scheduled on/off times. By providing real-time feedback, **WattWise** encourages energy-efficient habits and promotes sustainability.

Goals/Objectives:

- Demonstrate hardware-software (C++) integration using Arduino technology.
- Monitor and display energy savings in real-time using an LCD screen.
- Utilize a real-time clock (DS1307 RTC module) to maintain accurate timing.
- Design and fabricate a functional 3D-printed house shaped keychain holder using SolidWorks.
- Provide users with visual feedback and rewards for energy-saving behaviors.

Benefits and Application:

- Helps reduce unnecessary electricity consumption.
- Prototype development for future smart energy management products.
- Energy conservation and sustainability initiatives.



System Architecture / Electronics

- Arduino Uno controls the system using time data from the DS1307 RTC Module.
- The Arduino automatically operates the LED and fan motor based on programmed schedules.
- Energy savings are displayed on the LCD screen, with contrast adjusted by a potentiometer.

Skills and Learning Outcomes from the SURE Program:

- Hands-on experience with Arduino, Raspberry Pi and embedded systems.
- Practical programming and debugging skills.
- 3D printing and prototyping of mechanical components.
- System Integration (hardware + software).
- Hardware configuration and wiring.
- Assembly and testing of complete systems.
- Reverse engineering and problem-solving.
- Teamwork and collaboration in project environments.
- Project planning and time management.
- Technical documentation and presentation (posters, reports).
- Troubleshooting and iterative design improvements.
- Exposure to real-world engineering challenges.

